



Beatriz Martins Gomes Silva

Data Science | Machine Learning | Data Analytics

Biomedical Engineering student at the Federal University of Uberlândia (UFU), with international academic experience at Polytech Marseille and hands-on Data Science experience at Expleo Group on Airbus Helicopters projects in France.

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📁 Portfolio_BMGs

TECHNICAL SKILLS

- **Programming & Data:** Python (Pandas, NumPy, Matplotlib, scikit-learn), SQL, R (tidyverse, caret, rpart), MATLAB/Simulink
- **Machine Learning:** Classification, Regression, Clustering, Decision Trees, Model Evaluation
- **Data Analytics:** Data Cleaning, Exploratory Data Analysis (EDA), Correlation Analysis, Descriptive Statistics, Data Visualization
- Data Visualization & Low-Code: Looker Studio, AppSheet
- **Web & Other Technologies:** Java, C, HTML, CSS, JavaScript

PROFESSIONAL EXPERIENCE

DATA SCIENCE INTERN | Expleo Group (Airbus Helicopters) - Vitrolles, France

06/2026 - 07/2026

- Developed interactive Looker Studio dashboards for the analysis of financial datasets and quality-related indicators.
- Built an internal AppSheet troubleshooting workflow, structuring operational data collection and improving request traceability.
- Translated operational requirements into structured data solutions, dashboards and digital workflows for internal users.

UNDERGRADUATE RESEARCHER - CONTROL SYSTEMS | LASEC - Uberlândia, Brazil

03/2025 - 08/2025

- Developed a data acquisition pipeline for an experimental aeropendulum using ESP32 and MPU9250 sensors, collecting PWM-angle time-series data through serial communication at 10 Hz.
- Performed dynamic system modeling and identification using experimental data and validated the mathematical model against measured responses, achieving an 81.39 % model fit.

UNDERGRADUATE RESEARCHER - DATA SCIENCE | GIM - Uberlândia, Brazil

03/2023 - 09/2024

- Developed a classification model using the Breast Cancer Wisconsin dataset and Decision Trees.
- Evaluated model performance through confusion-matrix and ROC-based analysis, identified three key geometric predictors and derived an interpretable classification rule.

PROJECTS

BREAST CANCER DIAGNOSTIC CLASSIFICATION IN R

- Reproducible EDA-to-evaluation workflow on 569 observations; Decision Tree baseline achieved 85.96 % test accuracy and ROC-AUC 0.8862, with malignant-class error analysis.

ENGINEERING PROJECT WORKLOAD & PLANNING AUTOMATION

- Built a synchronized project-management system for workload, weekly time, progress, planning and KPI dashboarding.

SERVICE REQUEST WORKFLOW AUTOMATION

- Built a web-based request workflow converting operational inputs into validated, traceable and analytics-ready records.

TOWER OF IRIS

- Developed a Hanoi-inspired browser puzzle with rule validation, 12 target configurations, three game modes, scoring and timed challenges.

EDUCATION

Polytech Marseille (amU)

2025 - 2026

Biomedical Engineering – International Exchange Program (Ranked 1st)

Federal University of Uberlândia (UFU)

2022 - Present

B.Sc. in Biomedical Engineering | **Expected Graduation: 08/2027**

CERTIFICATIONS

R FOR DATA SCIENCE: DATA ANALYSIS AND STATISTICAL MODELING – 20 HOURS

2023

BIOMEDICAL SIGNAL PROCESSING AND ANALYSIS WITH R – 30 HOURS

2024

LANGUAGES

- Portuguese: Native
- French: Fluent (C2)
- English: Advanced (C1)